

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	Aim : Generate Histogram of Images, apply Histogram Equalization and Histogram Matching.	
Open Ended Problem 8	Date: 22/08/2026	Enrolment No:92400133050

GitHub Link :

Objective

To apply histogram equalization and histogram matching to the same set of degraded images used in OPE7, compare their results, and study how both techniques improve image contrast, brightness, and clarity.

1. Histogram Equalization

Histogram equalization is an image enhancement technique that improves image contrast by redistributing the pixel intensity values. It is useful when the intensity values of an image are concentrated in a limited or uneven range.

The main effects of histogram equalization are:

- **Improves overall image contrast.**
- **Redistributes pixel intensity values.**
- **Makes hidden details more visible.**
- **Improves the visual appearance of the image.**
- **Makes the image easier to analyze.**

This is consistent with the OPE7 description, which states that histogram equalization redistributes intensity values and can reveal details that are not clearly visible in the original image.

2. Histogram Matching

Histogram matching, also called histogram specification, changes the intensity distribution of an input image so that its histogram becomes similar to a selected reference image.

Unlike histogram equalization, which automatically produces a more balanced intensity distribution, histogram matching allows us to control the desired appearance by selecting a suitable reference image.

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The basic process is:

1. Read the degraded image.
2. Select a suitable reference image.
3. Calculate the histogram and cumulative distribution function (CDF) of the input image.
4. Calculate the histogram and CDF of the reference image.
5. Map the input intensity values to the reference intensity values.
6. Generate the histogram-matched image.
7. Compare the original, equalized, and matched images.

Code

```
import cv2

import numpy as np

import matplotlib.pyplot as plt

# =====

# IMAGE PATHS - SAME AS OPE 7

# =====

image_path_1 = '/content/1.jpeg'
image_path_2 = '/content/2.jpeg'
image_path_3 = '/content/3.jpeg'
image_path_4 = '/content/4.jpeg'
image_path_5 = '/content/5.jpeg'

# =====

# READ IMAGES IN GRAYSCALE

# =====
```

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```

src_image_1 = cv2.imread(image_path_1, cv2.IMREAD_GRAYSCALE)

src_image_2 = cv2.imread(image_path_2, cv2.IMREAD_GRAYSCALE)

src_image_3 = cv2.imread(image_path_3, cv2.IMREAD_GRAYSCALE)

src_image_4 = cv2.imread(image_path_4, cv2.IMREAD_GRAYSCALE)

src_image_5 = cv2.imread(image_path_5, cv2.IMREAD_GRAYSCALE)


images = [

    src_image_1,

    src_image_2,

    src_image_3,

    src_image_4,

    src_image_5

]


# =====
# CHECK IF IMAGES ARE LOADED
# =====


for i, image in enumerate(images, start=1):

    if image is None:

        print(f"Image {i} not found!")

        exit()


print("All 5 images loaded successfully!")


# =====
# 1. HISTOGRAM EQUALIZATION
# =====


equalized_images = []

```

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for image in images:

```
equalized = cv2.equalizeHist(image)
```

```
equalized_images.append(equalized)
```

```
# =====
```

```
# 2. HISTOGRAM MATCHING
```

```
# =====
```

```
# Image 1 is used as the reference image
```

```
reference_image = src_image_1
```

```
def histogram_matching(source, reference):
```

```
# -----
```

```
# Calculate histogram of source image
```

```
# -----
```

```
source_hist = cv2.calcHist(
```

```
    [source],
```

```
    [0],
```

```
    None,
```

```
    [256],
```

```
    [0, 256]
```

```
)
```

```
# -----
```

```
# Calculate histogram of reference image
```

```
# -----
```

```
reference_hist = cv2.calcHist(
```

```
    [reference],
```

```
    [0],
```

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```

None,

[256],

[0, 256]

)

# -----

# Calculate CDF of source and reference

# -----

source_cdf = source_hist.cumsum()

reference_cdf = reference_hist.cumsum()

# -----

# Normalize CDF

# -----

source_cdf = source_cdf / source_cdf[-1]

reference_cdf = reference_cdf / reference_cdf[-1]

# -----

# Create intensity mapping

# -----

mapping = np.zeros(256, dtype=np.uint8)

for source_value in range(256):

    difference = np.abs(

        reference_cdf - source_cdf[source_value]

    )

    mapping[source_value] = np.argmin(difference)

```

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```

# -----

# Apply mapping to source image

# -----

matched_image = mapping[source]

return matched_image

# =====
# APPLY HISTOGRAM MATCHING TO ALL IMAGES
# =====

matched_images = []

for image in images:

    matched = histogram_matching(
        image,
        reference_image
    )

    matched_images.append(matched)

# =====
# DISPLAY ORIGINAL, HISTOGRAM EQUALIZATION,
# AND HISTOGRAM MATCHING
# =====

plt.figure(figsize=(15, 20))

for i in range(5):

```

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```
# -----
```

```
# Original Image
```

```
# -----
```

```
plt.subplot(5, 3, i * 3 + 1)
```

```
plt.imshow(
    images[i],
    cmap='gray'
)
```

```
plt.title(
    f"Image {i+1} - Original"
)
```

```
plt.axis('off')
```

```
# -----
```

```
# Histogram Equalization
```

```
# -----
```

```
plt.subplot(5, 3, i * 3 + 2)
```

```
plt.imshow(
    equalized_images[i],
    cmap='gray'
)
```

```
plt.title(
    f"Image {i+1} - Histogram Equalization"
)
```

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```
plt.axis('off')
```

```
# -----
```

```
# Histogram Matching
```

```
# -----
```

```
plt.subplot(5, 3, i * 3 + 3)
```

```
plt.imshow(
    matched_images[i],
    cmap='gray'
)
```

```
plt.title(
    f"Image {i+1} - Histogram Matching"
)
```

```
plt.axis('off')
```

```
plt.tight_layout()
```

```
plt.show()
```

```
# =====
```

```
# DISPLAY HISTOGRAMS
```

```
# =====
```

```
for i in range(5):
```

```
plt.figure(figsize=(15, 4))
```

```
# -----
```

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```
# Original Histogram
```

```
# -----
```

```
plt.subplot(1, 3, 1)
```

```
plt.hist(
    images[i].ravel(),
    bins=256,
    range=[0, 256]
)
```

```
plt.title(
    f"Image {i+1} - Original Histogram"
)
```

```
plt.xlabel("Pixel Intensity")
plt.ylabel("Frequency")
```

```
# -----
```

```
# Equalized Histogram
```

```
# -----
```

```
plt.subplot(1, 3, 2)
```

```
plt.hist(
    equalized_images[i].ravel(),
    bins=256,
    range=[0, 256]
)
```

```
plt.title(
    f"Image {i+1} - Equalized Histogram"
)
```

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)

plt.xlabel("Pixel Intensity")

plt.ylabel("Frequency")

Matched Histogram

plt.subplot(1, 3, 3)

plt.hist(

 matched_images[i].ravel(),

 bins=256,

 range=[0, 256]

)

plt.title(

 f"Image {i+1} - Matched Histogram"

)

plt.xlabel("Pixel Intensity")

plt.ylabel("Frequency")

plt.tight_layout()

plt.show()

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Output

Image 1 - Original



Image 1 - Histogram Equalization



Image 1 - Histogram Matching



Image 2 - Original

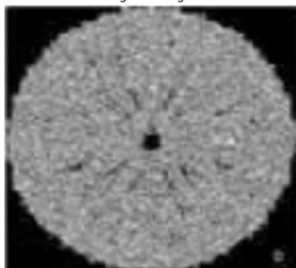


Image 2 - Histogram Equalization

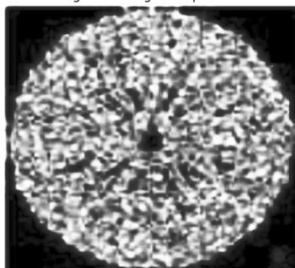


Image 2 - Histogram Matching

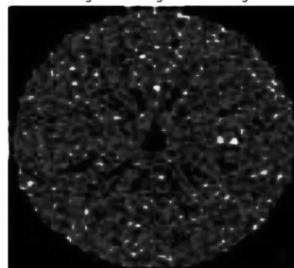


Image 3 - Original



Image 3 - Histogram Equalization



Image 3 - Histogram Matching



Image 4 - Original

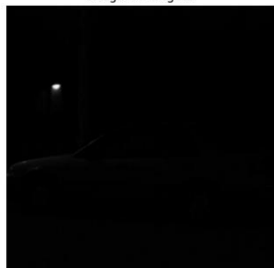


Image 4 - Histogram Equalization



Image 4 - Histogram Matching



Image 5 - Original

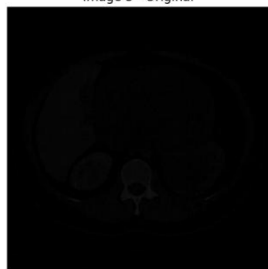
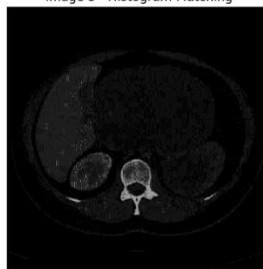


Image 5 - Histogram Equalization



Image 5 - Histogram Matching





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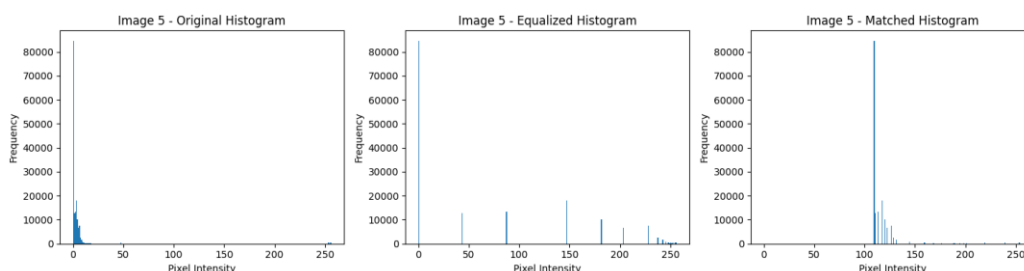
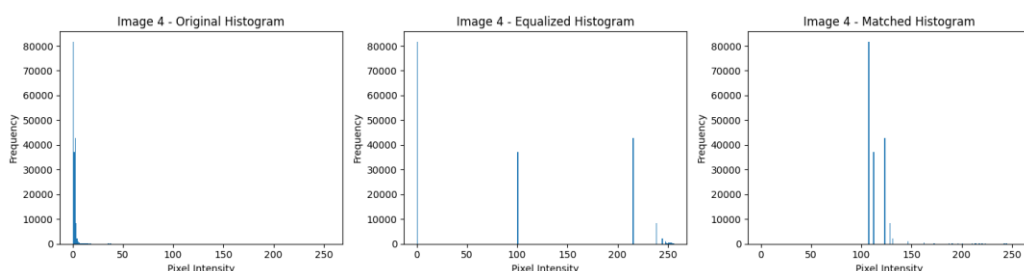
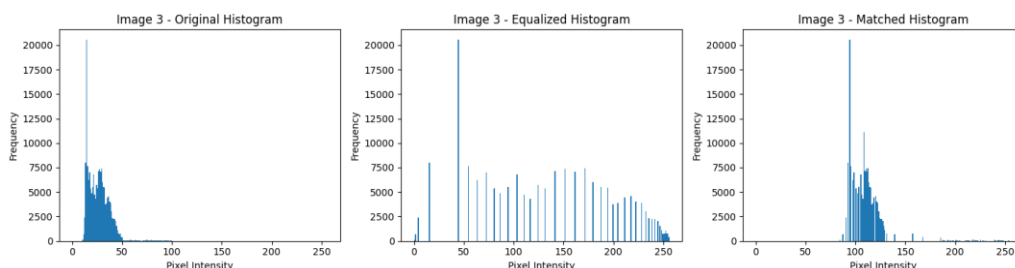
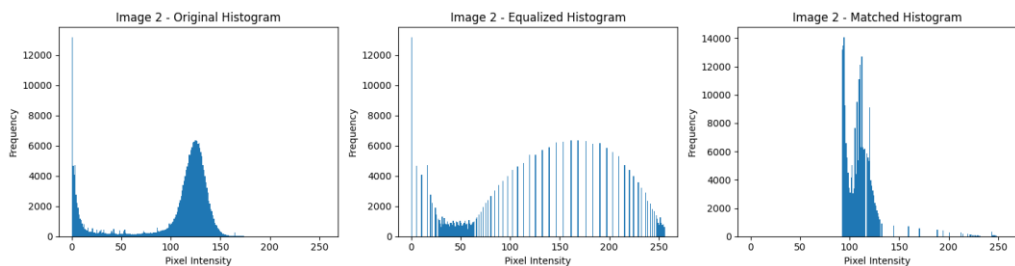
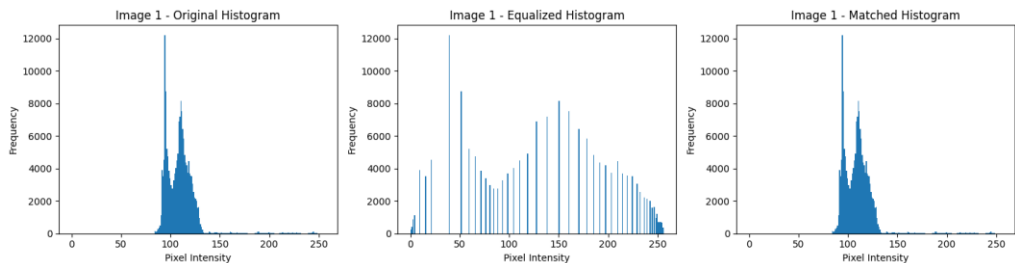
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Comparison

Histogram equalization is automatic and simple. It does not require a reference image and is effective for improving the overall contrast of degraded images. However, it can sometimes produce excessive contrast or an unnatural appearance. The OPE7 also notes that histogram equalization may sometimes over-enhance an image.

Histogram matching provides greater control because the desired intensity distribution is obtained from a reference image. Therefore, it can produce a more predictable appearance when an appropriate reference image is selected.

Effect on Contrast

Histogram Equalization: Generally increases the overall contrast by spreading and redistributing intensity values.

Histogram Matching: Changes the contrast according to the reference histogram.

Equalization is better for automatic enhancement.

Matching is better when a specific contrast appearance is required.

Effect on Clarity

Equalization can make hidden details more visible.

Matching can improve clarity while maintaining an intensity distribution similar to the reference.

A poor reference image can make histogram matching less effective.

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Conclusion

Both histogram equalization and histogram matching are useful techniques for improving degraded images.

Histogram equalization is simple, automatic, and effective for increasing overall contrast. It is suitable when no reference image is available and the main goal is to make the image clearer.

Histogram matching provides more control over the final image because the intensity distribution is based on a reference image. When a suitable reference image is available, histogram matching can produce a more natural and controlled result.

Therefore, histogram equalization is generally better for simple overall contrast enhancement, while histogram matching is better when a specific brightness and contrast appearance is required. The final performance depends on the characteristics of each degraded image and the quality of the selected reference image.